

Data Architecture & Governance

Data objects, ownership, lineage, and governance

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1. Introduction

This document defines the data architecture and governance framework for ArchiSurance. It catalogs all data objects, their ownership, lineage from source to consumption, and the governance policies that ensure data quality and compliance. The Application Portfolio (EA-APP-002) describes which systems access each data object. The Compliance and Regulatory Requirements document (EA-CMP-016) details the regulatory constraints affecting data handling.

The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. The data architecture governance has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023. In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel. Change management activities include stakeholder communication, training sessions, and documentation updates. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework. The architectural principles that guide this work include separation of concerns, loose coupling between components, preference for open standards, and the ability to scale horizontally under peak load conditions such as end-of-year policy renewals. ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns. Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility. The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone. The operational support model is defined in the Operational Runbook and includes on-call rotation and escalation procedures. Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register. Integration testing spans three environments: development, staging, and pre-production. Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. Implementation follows an iterative approach, with quarterly milestones aligned to the overall program timeline. Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys. The non-functional requirements specify a maximum end-to-end latency of five hundred milliseconds at the ninety-ninth percentile, measured under a sustained load of one thousand concurrent users. The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael Nygard, stored alongside the codebase in the version control system.

Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean

time to recovery, and user satisfaction scores collected through quarterly surveys. From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices. The vendor selection process followed the procurement guidelines established by the CFO's office. Implementation follows an iterative approach, with quarterly milestones aligned to the overall program timeline. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period. Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer.

Risk mitigation strategies include regular checkpoint reviews, automated regression testing, and staged rollout procedures. The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal. The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility. The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline.

The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. The total cost of ownership analysis, prepared by the Finance Department in collaboration with IT, indicated a payback period of approximately eighteen months assuming current utilization rates. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. The configuration management database maintains a record of all IT assets, including hardware serial numbers, software license keys, patch levels, and relationships between configuration items.

2. Data Object Catalog

2.x Policy Record

The Policy Record is a core data asset within the ArchiSurance data model. It is accessed by: Policy Admin, Data Warehouse.

Classification: PII. Retention period: 5 years. Backup frequency: Daily incremental, weekly full.

The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios. Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch via the CI/CD pipeline. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development.

2.x Claim Record

The Claim Record is a core data asset within the ArchiSurance data model. It is accessed by: CMP, Enterprise Data Warehouse.

Classification: Internal. Retention period: 5 years. Backup frequency: Daily incremental, weekly full.

Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period. An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog.

2.x Customer Record

The Customer Record is a core data asset within the ArchiSurance data model. It is accessed by: CRM System.

Classification: PII. Retention period: 5 years. Backup frequency: Daily incremental, weekly full.

The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam.

2.x Premium Calculation Data

The Premium Calculation Data is a core data asset within the ArchiSurance data model. It is accessed by: Risk Engine.

Classification: Internal. Retention period: 5 years. Backup frequency: Daily incremental, weekly full.

The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. Integration testing spans three environments: development, staging, and pre-production.

2.x Fraud Score Dataset

The Fraud Score Dataset is a core data asset within the ArchiSurance data model. It is accessed by: FDE.

Classification: Internal. Retention period: 5 years. Backup frequency: Daily incremental, weekly full.

The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points.

2.x Commission Ledger

The Commission Ledger is a core data asset within the ArchiSurance data model. It is accessed by: Billing Engine.

Classification: Internal. Retention period: 5 years. Backup frequency: Daily incremental, weekly full.

The vendor selection process followed the procurement guidelines established by the CFO's office. Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings.

2.x Audit Log

The Audit Log is a core data asset within the ArchiSurance data model. It is accessed by: Okta IAM.

Classification: Internal. Retention period: 7 years. Backup frequency: Daily incremental, weekly full.

Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices. The total cost of ownership analysis, prepared by the Finance Department in collaboration with IT, indicated a payback period of approximately eighteen months assuming current utilization rates. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership.

2.x Regulatory Submission Record

The Regulatory Submission Record is a core data asset within the ArchiSurance data model. It is accessed by: Cognos Analytics.

Classification: Internal. Retention period: 7 years. Backup frequency: Daily incremental, weekly full.

Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. A proof-of-concept was conducted over a four-week period, involving representatives from the Claims Department, IT Operations, and the Data Analytics Team to validate the technical feasibility. From a historical perspective, ArchiSurance first recognized the need for modernization in this area following an internal efficiency review conducted in late 2022, which highlighted significant manual effort and process duplication across regional offices. Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024.

2.x Document Metadata

The Document Metadata is a core data asset within the ArchiSurance data model. It is accessed by: Document Management System.

Classification: Internal. Retention period: 5 years. Backup frequency: Daily incremental, weekly full.

Scalability testing performed in the staging environment confirmed that the system handles a three-times-peak load without degradation, providing sufficient headroom for seasonal demand fluctuations. An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework. The solution was designed to be modular, allowing individual components to be upgraded or replaced without affecting the overall system availability or data integrity. The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes.

2.x Analytics Cube

The Analytics Cube is a core data asset within the ArchiSurance data model. It is accessed by: BI Platform, Oracle DWH.

Classification: Internal. Retention period: 5 years. Backup frequency: Daily incremental, weekly full.

Lessons learned from the pilot phase have been incorporated into the current design iteration. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael Nygard, stored alongside the codebase in the version control system.

3. Data Ownership and Stewardship

The Data Steward role is responsible for ensuring data quality standards are met. The Data Analytics Team manages the analytical data pipeline, while the Compliance Officer oversees regulatory data requirements.

Data Object	Business Owner	Data Steward	Classification
Policy Record	Policy Admin	L. Hendriks	PII
Claim Record	Claims Dept.	T. Mulder	PII
Customer Record	Sales Division	S. Visser	PII
Premium Calc Data	Underwriting	K. Bakker	Internal
Fraud Score Dataset	Data Analytics	R. de Jong	Confidential
Commission Ledger	Finance Dept.	M. de Vries	Internal
Audit Log	IT Operations	P. Jansen	Internal
Regulatory Sub.	Legal & Compl.	R. de Jong	Regulatory
Document Metadata	Policy Admin	L. Hendriks	Internal
Analytics Cube	Data Analytics	K. Bakker	Internal

4. Data Lineage

Data flows from source systems (CRM, Guidewire PolicyCenter, Claims Processing System) through the MuleSoft ESB to the Oracle DWH via the Batch Processing Engine. The Reporting & Analytics Platform consumes data from the warehouse for dashboards and regulatory reports.

ArchiSurance participates in an industry-wide information sharing and analysis center for insurance companies, contributing anonymized threat intelligence data and receiving early warnings about sector-specific attack patterns. Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. Stakeholders across multiple departments have contributed requirements for this workstream. The architectural principles that guide this work include separation of concerns, loose coupling between components, preference for open standards, and the ability to scale horizontally under peak load conditions such as end-of-year policy renewals. The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions. The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024. The data lineage traceability has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023. Post-implementation review sessions are scheduled for thirty and ninety days after go-live to capture lessons learned and identify optimization opportunities. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. Business continuity considerations were incorporated from the outset, with the DR Recovery Node in the secondary data center providing failover capacity; the most recent failover test was conducted in October 2024 and achieved recovery within the target RTO of one hour. Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch via the CI/CD pipeline. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer. The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership.

The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. The data retention policy mandates that transactional records are kept for seven years in online storage and archived to cold storage thereafter, with the ability to retrieve archived records within twenty-four hours upon regulatory request. The network team conducted a bandwidth utilization study in February 2025, confirming that inter-datacenter traffic remains below sixty percent of the available WAN capacity under normal operating conditions. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael Nygard, stored alongside the codebase in the version control system.

The vendor selection process followed the procurement guidelines established by the CFO's office. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. A chaos engineering program was introduced in Q1 2025, starting with controlled fault injection in the staging environment to validate the resilience of the system against network partitions and dependency failures. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios. The transition plan was developed in alignment with the enterprise change management framework, incorporating staged rollouts, parallel-run periods, and defined go/no-go criteria at each milestone. The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and

business processes. Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024.

Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point. The project governance structure includes a steering committee meeting monthly, a working group meeting weekly, and daily stand-ups during active development sprints. Data privacy impact assessment was filed with the Data Protection Officer in Q1 2025. A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline. The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes. Integration testing spans three environments: development, staging, and pre-production.

5. Data Storage Architecture

Operational data is stored in PostgreSQL DB (primary OLTP) and Oracle Database 19c (legacy billing and DWH). The Cognos Analytics uses Microsoft SQL Server 2022 for OLAP workloads. the search engine cluster indexes document metadata, and Redis provides caching for frequently accessed data.

Claims Database Schema (PostgreSQL)

```
CREATE TABLE claims (  
  claim_id          UUID PRIMARY KEY DEFAULT gen_random_uuid(),  
  policy_id         UUID NOT NULL REFERENCES policies(policy_id),  
  customer_id       UUID NOT NULL REFERENCES customers(customer_id),  
  claim_type        VARCHAR(50) NOT NULL,  
  claim_status      VARCHAR(30) NOT NULL DEFAULT 'REGISTERED',  
  amount_claimed    DECIMAL(12,2),  
  amount_approved   DECIMAL(12,2),  
  fraud_score       DECIMAL(5,4),  
  submitted_at      TIMESTAMPTZ NOT NULL DEFAULT now(),  
  assessed_at       TIMESTAMPTZ,  
  settled_at        TIMESTAMPTZ,  
  handler_id        UUID REFERENCES employees(employee_id),  
  CONSTRAINT chk_status CHECK (claim_status IN  
    ('REGISTERED','ASSESSING','APPROVED','REJECTED','FRAUD_REVIEW','SETTLED'))  
);  
  
CREATE INDEX idx_claims_policy ON claims(policy_id);  
CREATE INDEX idx_claims_customer ON claims(customer_id);  
CREATE INDEX idx_claims_status ON claims(claim_status);  
CREATE INDEX idx_claims_submitted ON claims(submitted_at);
```

Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. Security review was completed in accordance with the CISO's security-by-design framework. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. The architecture team conducted a threat modeling exercise using the STRIDE framework, identifying potential spoofing, tampering, and information disclosure risks; mitigations for each threat were documented and tracked in the security risk register. Integration testing spans three environments: development, staging, and pre-production. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service

Manager and the on-call engineer. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. The information security team requires that all new systems undergo a penetration test before production deployment, with findings above medium severity remediated prior to go-live approval. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. The solution was designed to be modular, allowing individual components to be upgraded or replaced without affecting the overall system availability or data integrity. Stakeholders across multiple departments have contributed requirements for this workstream. The enterprise architecture repository in Sparx EA is the authoritative source for all model elements; any changes to this document must be reflected there within five business days. The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. The configuration management database maintains a record of all IT assets, including hardware serial numbers, software license keys, patch levels, and relationships between configuration items. A chaos engineering program was introduced in Q1 2025, starting with controlled fault injection in the staging environment to validate the resilience of the system against network partitions and dependency failures. Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. The vendor selection process followed the procurement guidelines established by the CFO's office. Regression test suites are maintained by the QA team and executed automatically on every merge to the main branch via the CI/CD pipeline. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline.

6. Data Quality Framework

ArchiSurance employs a data quality framework with automated checks running daily. Quality dimensions measured include: completeness, accuracy, consistency, timeliness, and uniqueness. Quality scores are published to the Analytics Dashboard weekly.

The vendor selection process followed the procurement guidelines established by the CFO's office. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. Integration testing spans three environments: development, staging, and pre-production. The platform engineering team provides a standardized deployment template that all application teams are required to use, ensuring consistency in logging, health checks, and resource limits. Security review was completed in accordance with the CISO's security-by-design framework. Cost allocation for shared infrastructure follows a chargeback model based on resource consumption metrics, with monthly reports sent to each business unit for transparency and accountability. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. Capacity planning exercises revealed that the current infrastructure can sustain projected growth for the next twelve months, after which additional compute and storage resources will be required. Deployment windows are scheduled every Tuesday and Thursday between 06:00 and 08:00 CET, with emergency deployments requiring approval from both the IT Service Manager and the on-call engineer. Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. The data retention policy mandates that transactional records are kept for seven years in online storage and archived to cold storage thereafter, with the ability to retrieve archived records within twenty-four hours upon regulatory request. The API design guidelines specify that all REST endpoints must support content negotiation, return standard HTTP status codes, include correlation identifiers in response headers, and provide machine-readable error bodies in RFC 7807 format. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently

in the Adopt or Trial rings. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period. Performance benchmarks are captured in the enterprise monitoring framework and reported on a monthly basis to leadership. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024. An internal audit in March 2025 confirmed that the controls implemented for this area meet the requirements of the enterprise risk management framework. Post-implementation review sessions are scheduled for thirty and ninety days after go-live to capture lessons learned and identify optimization opportunities. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. A formal lessons-learned workshop is held at the end of each quarter, with outcomes published to the internal wiki and action items assigned to specific team members with target completion dates. Business continuity considerations were incorporated from the outset, with the DR Recovery Node in the secondary data center providing failover capacity; the most recent failover test was conducted in October 2024 and achieved recovery within the target RTO of one hour. The operational acceptance criteria include automated smoke tests, performance baselines, runbook completeness, and sign-off from both the IT Service Manager and the business process owner. The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes. Application health is assessed using a synthetic monitoring approach, where automated scripts simulate real user journeys every five minutes and report availability from three geographically distributed probe locations. The data quality framework has been a critical area of focus for ArchiSurance since the digital transformation initiative began in 2023. The development team follows trunk-based development with short-lived feature branches, automated linting, and a code coverage threshold of eighty percent enforced by the CI pipeline.

7. Data Governance Policies

Data governance at ArchiSurance is driven by regulatory requirements including GDPR Compliance and SOX Audit Trail. The governance council meets monthly to review data quality metrics and policy exceptions.

Lessons learned from the pilot phase have been incorporated into the current design iteration. Security review was completed in accordance with the CISO's security-by-design framework. The vendor risk assessment framework evaluates suppliers across five dimensions: financial stability, technology roadmap alignment, support quality, security posture, and contractual flexibility. The architecture review confirmed that the proposed solution aligns with the TOGAF Architecture Development Method phases used by ArchiSurance, specifically phases B through D covering Business, Information Systems, and Technology Architecture. Monitoring and alerting are integrated with the central observability stack based on Prometheus and Grafana. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam. In the context of insurance operations, peak load occurs during natural disaster events when claim submissions can spike by up to ten times the normal rate; the auto-scaling configuration accounts for this burst pattern. The operational support model is defined in the Operational Runbook and includes on-call rotation and escalation procedures. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. The architecture team maintains a dependency register that maps each component to its upstream providers and downstream consumers, updated quarterly. Dependencies on third-party services were documented and risk-assessed, with fallback mechanisms defined for each critical integration point. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. A dedicated Slack channel was established for real-time coordination between development, operations, and security teams during the rollout period. Service-level indicators are collected continuously and aggregated into dashboards visible to both technical teams and business stakeholders. An accessibility audit was conducted by an external specialist in January 2025 to ensure that all user-facing components meet WCAG 2.1 Level AA standards, with remediation items tracked in the product backlog. Cross-functional workshops held in January and March 2025 helped refine the scope and integration points. The business process documentation follows the BPMN 2.0 standard, with process models maintained in the Signavio process modeling tool and linked to the ArchiMate model elements in Sparx EA. The enterprise technology radar, published semi-annually by the CTO's office, classifies technologies as Adopt, Trial, Assess, or Hold; the technologies used in this area are all currently in the Adopt or Trial rings. The enterprise identity provider federates with the national eHerkenning system for business-to-government

interactions and with DigiD for customer-facing authentication on the self-service portal. Documentation standards require that all architecture decisions are recorded as Architecture Decision Records using the ADR format proposed by Michael Nygard, stored alongside the codebase in the version control system. The architecture decisions were reviewed and approved by the Architecture Review Board in Q2 2024. The non-functional requirements specify a maximum end-to-end latency of five hundred milliseconds at the ninety-ninth percentile, measured under a sustained load of one thousand concurrent users. Automated deployment pipelines leverage the CI/CD infrastructure described in the platform engineering documentation. The total cost of ownership analysis, prepared by the Finance Department in collaboration with IT, indicated a payback period of approximately eighteen months assuming current utilization rates. Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. A comprehensive training program was developed for end users, including online modules, instructor-led sessions, and a sandbox environment for hands-on practice.

Security review was completed in accordance with the CISO's security-by-design framework. External consultants from Deloitte provided an independent assessment of the architecture in late 2024. The design considers both current-state constraints and target-state aspirations as documented in the strategic roadmap. Stakeholders across multiple departments have contributed requirements for this workstream. The vendor selection process followed the procurement guidelines established by the CFO's office. The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes. Collaboration with the vendor support team has been essential during the implementation phase; weekly technical syncs ensure that configuration changes are validated against the vendor's reference architecture before being applied to the production environment. The solution leverages event-driven patterns wherever possible, with Apache Kafka providing durable, ordered event streams that decouple producers from consumers and enable replay for debugging and reprocessing scenarios. The team uses a definition of done that includes code review by at least two peers, passing CI pipeline, updated architecture diagrams, and entry in the change log maintained by the Release Manager. The data classification exercise, completed in Q4 2024, assigned sensitivity labels to all data objects, informing encryption, access control, and retention policies.

Lessons learned from the pilot phase have been incorporated into the current design iteration. Feedback from the user acceptance testing phase indicated that the new interface reduced the average task completion time by approximately thirty percent compared to the legacy system, though some users requested additional keyboard shortcuts for power-user workflows. The enterprise data model, maintained by the Data Architecture team, defines canonical schemas for all shared data objects; integrating systems are expected to conform to these schemas or implement transformation layers that map proprietary formats to the canonical model. The total cost of ownership analysis, prepared by the Finance Department in collaboration with IT, indicated a payback period of approximately eighteen months assuming current utilization rates. Security review was completed in accordance with the CISO's security-by-design framework. Key performance indicators for the initiative include transaction throughput, error rate, mean time to detection, mean time to recovery, and user satisfaction scores collected through quarterly surveys. The enterprise identity provider federates with the national eHerkenning system for business-to-government interactions and with DigiD for customer-facing authentication on the self-service portal. The observability strategy follows the three pillars model: structured logs shipped to Elasticsearch, distributed traces collected via OpenTelemetry, and metrics scraped by Prometheus; all three are unified in Grafana dashboards. The data classification exercise, completed in Q4 2024, assigned sensitivity labels to all data objects, informing encryption, access control, and retention policies. Interoperability with partner systems is achieved through the Partner Integration Hub, which exposes a curated subset of the internal API surface with additional rate limiting, schema validation, and audit logging applied at the gateway layer.

Capacity forecasting models project a twenty-five percent year-over-year increase in data volume, driven primarily by the digitization of paper-based processes and the expansion of the broker channel. Technical debt in this area was cataloged during a dedicated sprint in Q3 2024; the resulting backlog of forty-seven items has been prioritized by business impact and is being addressed alongside feature development. The Architecture Review Board convened a special session in February 2025 to evaluate the implications of the proposed changes on downstream systems and business processes. Security review was completed in accordance with the CISO's security-by-design framework. Data

privacy impact assessment was filed with the Data Protection Officer in Q1 2025. Budget allocation was approved as part of the annual IT investment portfolio review conducted in November 2024. During the initial assessment phase, the team identified several areas where the existing architecture fell short of the target-state requirements defined in the enterprise roadmap. The initiative is tracked in the enterprise PMO tool with bi-weekly status updates to the steering committee. The incident management process defines four severity levels, with Severity 1 incidents requiring a response within fifteen minutes and a post-incident review within five business days. The enterprise backup strategy uses a 3-2-1 approach: three copies of data, on two different media types, with one copy stored offsite at the secondary data center in Rotterdam.

8. Appendices

References and Related Documents

Doc ID	Title	Version
EA-OVR-001	Enterprise Architecture Overview	3.0
EA-APP-002	Application Portfolio & Inventory	4.2
EA-INFRA-003	Infrastructure Design Document	2.5
EA-BIZ-004	CRM Business Process Description	2.0
EA-CLM-005	Claims Processing Workflow	3.1
EA-SEC-006	Security Architecture	2.3
EA-INT-007	Integration Architecture	2.8
EA-DAT-008	Data Architecture & Governance	2.0
EA-CLD-009	Cloud Migration Assessment	1.5
EA-DR-010	Disaster Recovery & Business Contin	3.0
EA-NET-011	Network Architecture	2.2
EA-ROAD-012	Strategic IT Roadmap 2025-2027	2.0
EA-SVC-013	IT Service Catalog	3.5
EA-VND-014	Vendor & Technology Assessment	1.8
EA-OPS-015	Operational Runbook: Core Platform	4.1
EA-CMP-016	Compliance & Regulatory Requirement	2.5
EA-CHG-017	Change Management: ERP Migration	1.2

Distribution List

Name	Role	Responsibility
J. van den Berg	Enterprise Architect	Review & Approve
M. de Vries	Application Architect	Review & Contribute
P. Jansen	Integration Architect	Review & Contribute
K. Bakker	Infrastructure Architect	Review & Contribute
S. Visser	Business Analyst	Inform
T. Mulder	Process Architect	Inform
R. de Jong	CISO	Review & Approve
L. Hendriks	Data Architect	Inform
A. Willems	CTO	Final Approval
B. Smit	Head of Claims	Business Sign-off

Appendix: Meeting Minutes - Architecture Review Board

Date: March 15, 2025. Location: ArchiSurance HQ, Room 4.12. Attendees: J. van den Berg (chair), M. de Vries, P. Jansen, K. Bakker, S. Visser, T. Mulder (remote). Apologies: R. de Jong (on leave).

Agenda item 1: Review of open action items from previous meeting. Action 2024-47 (Jansen): Provide updated cost estimate for Oracle migration. Status: Completed. The revised estimate is EUR 1.4M including contingency. Action 2024-48 (de Vries): Schedule penetration test with external vendor. Status: In progress, vendor shortlist reduced to two candidates.

Agenda item 2: Discussion on coffee machine replacement in the 3rd floor kitchen. The current Jura E8 has been experiencing frequent bean grinder failures. Facilities will procure a replacement by end of month. Not related to architecture but raised due to impact on team productivity during long review sessions.

Agenda item 3: Parking lot discussion about whether to adopt Backstage as the internal developer portal. Deferred to next meeting pending PoC results. K. Bakker noted that the onboarding documentation is outdated and should be

refreshed regardless of the portal decision.

Next meeting: April 12, 2025 at 14:00 CET. Van den Berg to circulate agenda by April 5.

Detailed Revision Notes for Version 2.1

Section 2.3 was updated to reflect the new naming convention adopted by the Platform Engineering team. All references to 'microservice cluster' have been replaced with 'application workload group' per the Architecture Board decision of January 2025. Section 4.1 was expanded to include the results of the capacity planning exercise conducted in Q4 2024, which identified the need for additional compute resources in the Amsterdam primary data center.

Appendix B was added to include the RACI matrix requested by the PMO during the Q3 2024 quarterly review. The glossary was updated to include twelve new acronyms that were introduced since the previous version. Figure 3 was replaced with an updated diagram generated from the Sparx EA repository on March 28, 2025. The document classification was elevated from 'Internal' to 'Internal - Confidential' following the data classification review of February 2025.

Feedback from the peer review conducted by M. de Vries on March 10, 2025, addressed three items: (1) clarification of the boundary between the DMZ and the application tier, (2) addition of the backup retention policy reference, and (3) correction of the SLA target for the Claims Handling Service from 99.5% to 99.9% as agreed in the contract amendment of December 2024.

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